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SIMATS Online Programming Lab
PYTHON PROGRAMMING

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Input: 12 5 18 3 25 9
output:
Smallest number:3
Largest number:25

Your Code

```
arr = list(map(int, input("Enter array elements: ").split()))
n = len(arr)
smallest = arr[0]
largest = arr[0]
for i in arr:
    if i < smallest:
        smallest = i
    if i > largest:
        largest = i
print("Smallest number : ", smallest)
print("Largest number : ", largest)
```

Input
12 5 18 3 25 9

Output
Enter array elements: Smallest number : 3
Largest number : 25

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Input: 4 2 1 0 3
output:3

Your Code

```
arr = list(map(int, input("Enter array elements: ").split()))
n = len(arr) + 1
total = n * (n + 1) // 2
sum_arr = sum(arr)
missing = total - sum_arr
print("Missing number: ", missing)
```

Input
4 2 1 0 3

Output
Enter array elements: Missing number: 3

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Input: 12 5 18 3 25 9
OUTPUT: 18

Your Code

```
arr = list(map(int, input("Enter array elements: ").split()))
largest = second = -999999
for i in arr:
    if i > largest:
        second = largest
        largest = i
    elif i > second and i < largest:
        second = i
print("Second largest number: ", second)
```

Input
12 5 18 3 25 9

Output
Enter array elements: Second largest number: 18

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WAP to print the most repeated number in an array
INPUT: 2 4 6 2 5 3 1 2
OUTPUT: 2

Your Code

```
arr = list(map(int, input("Enter array elements: ").split()))
max_count = 0
most = arr[0]
for i in arr:
    count = 0
    for j in arr:
        if i == j:
            count += 1
    if count > max_count:
        max_count = count
        most = i
print("Most repeated number: ", most)
```

Input
2 4 6 2 5 3 1 2

Output
Enter array elements: Most repeated number: 2

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Q6 Unique elements 10 marks

WAP to print all unique elements in an array
INPUT: 2 4 6 2 5 6 7
OUTPUT: 4 5 7

Your Code

```
a = list(map(int, input().split()))
for i in range(len(a)):
    found = False
    for j in range(i + 1, len(a)):
        if a[i] == a[j]:
            found = True
            break
    if not found:
        print(a[i], end=" ")
```

Input
2 4 6 2 5 6 7

Output
2 4 6 5 7

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Q6 Rotate an array by 'N' positions 10 marks

WAP to rotate an array by 'N' positions
INPUT: 2 5 6 1 3 4
OUTPUT: 6 1 3 4 2 5

Your Code

```
n = list(map(int, input().split()))
for i in range(2, len(n)):
    print(n[i], end=" ")
for i in range(2):
    print(n[i], end=" ")
```

Input
2 5 6 1 3 4

Output
6 1 3 4 2 5

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Q7 Remove all duplicates in an array 10 marks

WAP to remove all duplicates in an array
INPUT: 2 4 6 2 7 8 3 4
OUTPUT: 2 4 6 7 8 3

Your Code

```
arr = list(map(int, input().split()))
for i in range(len(arr)):
    duplicate = False
    for j in range(i + 1, len(arr)):
        if arr[i] == arr[j]:
            duplicate = True
            break
    if not duplicate:
        print(arr[i], end=" ")
```

Input
2 4 6 2 7 8 3 4

Output
2 4 6 7 8 3

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Q8 Print all 0's followed by non-zero without changing the order 10 marks

WAP to print all 0's followed by non-zero without changing the order
INPUT: 1 0 4 0 2 5 0 7 0 6 0
OUTPUT: 0 0 0 0 0 1 4 2 5 7 6

Your Code

```
a = list(map(int, input().split()))
b = []
for i in a:
    if i == 0:
        b.append(i)
    else:
        b.append(i)
for i in b:
    print(i, end=" ")
```

Input
1 0 4 0 2 5 0 7 0 6 0

Output
0 0 0 0 0 1 4 2 5 7 6

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Consonants:7
Digits:3
Spaces:2

Your Code

```
s = input()
v = 0
d = 0
sp = 0
for ch in s:
    if ch in "aeiouAEIOU":
        v += 1
    elif ("0" <= ch <="9") or ("a" <= ch <="z"):
        d += 1
    elif " " == ch:
        sp += 1
    else:
        d += 1
```

Input
Hello World 123

Output
Vowels:3
Consonants:7
Digits:3
Spaces:2

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Your Code

```
s = input()
n = len(s)
for i in range(n // 2):
    rev = s[n - i - 1]
    if s[i] == rev:
        print(i, "is a palindrome")
    else:
        print(i, "is not a palindrome")
```

Input
malayalam

Output
malayalam is a palindrome

Visible Test Cases

- Test Case 1 ☒
- Test Case 2 ☒

10.00 / 10 marks